

Meiosis, Karyotype, Chromosomal Mutation Review

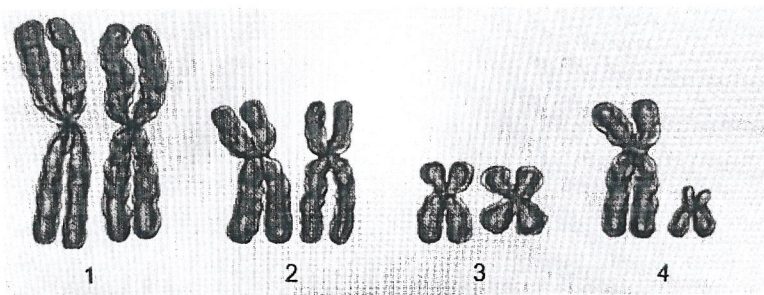
Key

For each missing word or phrase, circle the choice that correctly completes the sentence.

1. A *Drosophila* fruit fly cell with $2N = 8$ chromosomes—two of each kind—is called a (~~haploid~~ / diploid) cell. *Drosophila* fruit flies produce gametes with one of each kind of chromosome. These gametes are called (~~haploid~~ / diploid) cells. These cells contain (0.5N / 1N / 2N / 4N) chromosomes.
2. How many sperm cells are formed from an original cell in a *Drosophila* fruit fly at the conclusion of meiosis?
 - a. two haploid sperm cells
 - b. two diploid sperm cells
 - c. four haploid sperm cells
 - d. four diploid sperm cells
3. Which event occurs in meiosis, but not mitosis?
 - a. Separated chromatids move to opposite ends of the cell.
 - b. All chromosomes align along the center of the cell.
 - c. Homologous chromosomes pair to form a tetrad.
 - d. The parent cell divides to form genetically-identical daughter cells.
4. A cell undergoes meiosis. How do the gametes it produces compare with the original cell and with one another?
 - a. The gametes are genetically identical with the parent cell and with one another.
 - b. The gametes have half the chromosomes of the parent cell, and are genetically identical with one another.
 - c. The gametes have half the chromosomes of the parent cell, and are genetically different from one another.
 - d. The gametes have one fourth of the chromosomes of the parent cell, and are genetically identical with one another.

Circle each choice that correctly completes the sentence.

5. An arrangement of all the chromosomes of a diploid cell is called a(n) (~~phenotype~~ / ~~genotype~~ / karyotype / ~~genome~~). For a human cell, this arrangement contains 22 pairs of (autosomes / sex chromosomes / ~~genomes~~ / ~~karyotypes~~) and one pair of (autosomes / sex chromosomes / ~~genomes~~ / ~~karyotypes~~).
6. The diagram shows four pairs of chromosomes from a human karyotype.



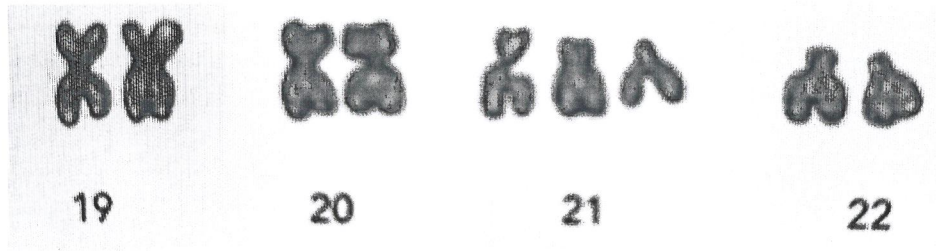
Which pair of chromosomes are the sex chromosomes?

- a. Pair 1
- b. Pair 2
- c. Pair 3
- d. Pair 4

For each missing word or phrase, circle the choice that correctly completes the sentence.

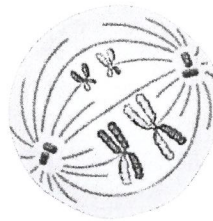
7. Gametes with an abnormal number of chromosomes may form as a result of (crossing over / point mutations / frameshift mutations / nondisjunction) during the process of (mitosis / meiosis / transcription / translation).

8. The diagram shows a section of the karyotype from a human donor. Chromosome pairs 19 through 22 are labeled.



Which statement describes the gametes that joined together to form this karyotype?

- a. The gametes formed normally, and then one chromosome was added when they joined together.
 - b. The gametes formed normally, and then one chromosome split apart when they joined together.
 - ☒ c. One gamete formed abnormally because a pair of homologous chromosomes failed to separate.
 - d. Both gametes formed abnormally because in each gamete, one pair of homologous chromosomes failed to separate.
9. The illustration shown represents which stage of meiosis?
- a. prophase I
 - b. anaphase II
 - c. telophase I
 - ☒ d. metaphase I



10. Explain your answer for what chose in question 9

Because the homologous chromosomes are side to side in the middle

11. Unlike mitosis, meiosis in male mammals results in the formation of
- a. one haploid gamete.
 - b. three diploid gametes.
 - c. four diploid gametes
 - ☒ d. four haploid gametes.
12. A normal body cell of a coyote has 78 chromosomes in its nucleus. What are the diploid (2N) and haploid (N) numbers of the coyote?

$$2N = 78 \quad N = 39$$

How many chromosomes would be in each coyote gamete?

39 chromosomes